

Exercise Series 3 (solution)

Ex1 (a) $\underline{y} \sim N_n(\underline{X}\underline{\theta}, \sigma^2 \mathbf{I})$

$$\therefore L(\underline{\theta}, \sigma^2) = \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \|\underline{y} - \underline{X}\underline{\theta}\|^2}$$

$$\ell(\underline{\theta}, \sigma^2) = \log L(\underline{\theta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \|\underline{y} - \underline{X}\underline{\theta}\|^2$$

Notice that $\underline{y} \sim N_n(\underline{X}\underline{\theta}, \sigma^2 \mathbf{I}) \Leftrightarrow y_i \overset{\text{indep}}{\sim} N(x_i^T \underline{\theta}, \sigma^2)$

$$\begin{aligned} \therefore L(\underline{\theta}, \sigma^2) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2\sigma^2} (y_i - \underline{x}_i^T \underline{\theta})^2} \\ &= \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \underline{x}_i^T \underline{\theta})^2} \\ &= \frac{1}{(2\pi)^{n/2} \sigma^n} e^{-\frac{1}{2\sigma^2} \|\underline{y} - \underline{X}\underline{\theta}\|^2} \end{aligned}$$

This is an alternative way of deriving the likelihood.

(b) MLE of $\underline{\theta}$ and σ^2 are obtained by maximizing $L(\underline{\theta}, \sigma^2)$, or equivalently, $\ell(\underline{\theta}, \sigma^2)$ w.r.t. $(\underline{\theta}, \sigma^2)$.

This can be done directly, by taking the partial derivatives and equating them with 0 to get feasible solutions, and by checking whether the Hessian is negative definite.

Alternatively, notice that for a fixed σ^2 , the log-likelihood is maximized when $\|\underline{y} - \underline{X}\underline{\theta}\|^2$ is minimized w.r.t. $\underline{\theta}$.

We have already seen that $\|\underline{y} - \underline{X}\underline{\theta}\|^2$ is min. w.r.t. $\underline{\theta}$ for $\underline{\theta} = \hat{\underline{\theta}}_{\text{LSE}} = (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{y}$.

For this choice of $\underline{\theta}$, the log-lik. as a function of σ^2 becomes $\ell(\sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\underline{y} - \underline{X}\hat{\underline{\theta}}_{\text{LSE}}\|^2$

Verify that $\ell(\sigma^2)$ is maximized at $\sigma^2 = \frac{\|\underline{y} - X\hat{\underline{\theta}}_{\text{LSE}}\|^2}{n}$

So, the MLE of $(\underline{\theta}, \sigma^2)$ is $(\hat{\underline{\theta}}_{\text{MLE}}, \hat{\sigma}_{\text{MLE}}^2)$, where

$$\hat{\underline{\theta}}_{\text{MLE}} = (X^T X)^{-1} X^T \underline{y}, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{\|\underline{y} - X\hat{\underline{\theta}}_{\text{MLE}}\|^2}{n}$$

(c) Clearly, $\hat{\underline{\theta}}_{\text{MLE}} = \hat{\underline{\theta}}_{\text{LSE}}$.

$$\text{Also, } \hat{\sigma}_{\text{MLE}}^2 = \frac{\|\underline{y} - X\hat{\underline{\theta}}_{\text{MLE}}\|^2}{n} = \frac{\|\underline{y} - X\hat{\underline{\theta}}_{\text{LSE}}\|^2}{n} = \frac{\|\hat{\underline{e}}\|^2}{n} = \frac{n-d}{n} \hat{\sigma}_{\text{LSE}}^2$$

(d) $\hat{\underline{\theta}}_{\text{MLE}} = \hat{\underline{\theta}}_{\text{LSE}} \Rightarrow E(\hat{\underline{\theta}}_{\text{MLE}}) = \underline{\theta}$ and $\text{Var}(\hat{\underline{\theta}}_{\text{MLE}}) = \sigma^2 (X^T X)^{-1}$

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{\|\underline{y} - X\hat{\underline{\theta}}_{\text{LSE}}\|^2}{n} = \frac{\|\hat{\underline{e}}\|^2}{n}$$

We have seen that $\|\hat{\underline{e}}\|^2 \sim \sigma^2 \chi_{n-d}^2$

$$\Rightarrow E(\|\hat{\underline{e}}\|^2) = \sigma^2 (n-d)$$

$$\text{Var}(\|\hat{\underline{e}}\|^2) = 2\sigma^4 (n-d)$$

$$\therefore E(\hat{\sigma}_{\text{MLE}}^2) = \frac{E(\|\hat{\underline{e}}\|^2)}{n} = \frac{n-d}{n} \sigma^2 = \left(1 - \frac{d}{n}\right) \sigma^2$$

$$\text{Var}(\hat{\sigma}_{\text{MLE}}^2) = \frac{\text{Var}(\|\hat{\underline{e}}\|^2)}{n^2} = \frac{2(n-d)}{n^2} \sigma^4 = 2\left(1 - \frac{d}{n}\right) \frac{\sigma^4}{n}$$

$\hat{\sigma}_{\text{MLE}}^2$ is a biased estimator of σ^2 .

Ex2 (a) $A = \sum_{i=1}^n \lambda_i \underline{u}_i \underline{u}_i^T$ (spectral decomposition)

$$\therefore \underline{z}^T A \underline{z} = \sum_{i=1}^n \lambda_i (\underline{u}_i^T \underline{z})^2 = \sum_{i=1}^n \lambda_i w_i^2$$

$$\begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} \underline{u}_1^T \underline{z} \\ \vdots \\ \underline{u}_n^T \underline{z} \end{pmatrix} = \begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_n^T \end{pmatrix} \underline{z} \sim N_n(\underline{0}, \sigma^2 I)$$

$$\therefore w_1, \dots, w_n \stackrel{\text{iid}}{\sim} N(0, 1)$$

$$(b) \quad E(\underline{z}^T A \underline{z}) = E\left(\sum_{i=1}^n \lambda_i w_i^2\right) = \sum_{i=1}^n \lambda_i E(w_i^2) = \sum_{i=1}^n \lambda_i$$

$(w_i \sim N(0,1) \Rightarrow w_i^2 \sim \chi_1^2)$

$$\text{Now, } A = \sum_{i=1}^n \lambda_i \underline{u}_i \underline{u}_i^T = U \Lambda U^T \Rightarrow \text{tr}(A) = \text{tr}(\Lambda) = \sum_{i=1}^n \lambda_i$$

$$\therefore E(\underline{z}^T A \underline{z}) = \text{tr}(A)$$

$$\begin{aligned} \underline{\text{Alt:}} \quad E(\underline{z}^T A \underline{z}) &= E \text{tr}(\underline{z}^T A \underline{z}) = E \text{tr}(A \underline{z} \underline{z}^T) \\ &= \text{tr}\{E(A \underline{z} \underline{z}^T)\} \\ &= \text{tr}\{A \underbrace{E(\underline{z} \underline{z}^T)}_I\} \quad \because E(\underline{z}) = \underline{0} \\ &= \text{tr}(A). \end{aligned}$$

$$\begin{aligned} \text{Var}(\underline{z}^T A \underline{z}) &= \text{Var}\left(\sum_{i=1}^n \lambda_i w_i^2\right) \\ &= \sum_{i=1}^n \lambda_i^2 \text{Var}(w_i^2) \quad \left[\because w_1, \dots, w_n \text{ are indep} \Rightarrow \text{Cov}(w_i^2, w_j^2) = 0 \text{ } \forall i \neq j\right] \\ &= 2 \sum_{i=1}^n \lambda_i^2 \quad (w_i^2 \sim \chi_1^2 \Rightarrow \text{Var}(w_i^2) = 2) \end{aligned}$$

$$\begin{aligned} A = U \Lambda U^T \Rightarrow A^2 &= U \Lambda^2 U^T \Rightarrow \text{tr}(A^2) = \text{tr}(\Lambda^2) = \sum_{i=1}^n \lambda_i^2 \\ \therefore \text{Var}(\underline{z}^T A \underline{z}) &= 2 \text{trace}(A^2). \end{aligned}$$

$$(c) \quad \underline{z} \sim N_n(\underline{0}, \Sigma), \quad \Sigma \succ 0. \text{ So, there exists } \Sigma^{1/2} \succeq 0 \text{ and } \underline{z}_0 \sim N_n(\underline{0}, I) \text{ s.t. } \underline{z} \stackrel{d}{=} \Sigma^{1/2} \underline{z}_0$$

$$\text{Now, } \underline{z}^T A \underline{z} = \underline{z}_0^T \Sigma^{1/2} A \Sigma^{1/2} \underline{z}_0$$

$$A \text{ p.s.d.} \Rightarrow \Sigma^{1/2} A \Sigma^{1/2} \text{ is p.s.d.}$$

$$\begin{aligned} \Rightarrow E(\underline{z}^T A \underline{z}) &= \text{trace}(\Sigma^{1/2} A \Sigma^{1/2}) \quad [\text{from part (b)}] \\ &= \text{trace}(A \Sigma^{1/2} \Sigma^{1/2}) = \text{trace}(A \Sigma) \end{aligned}$$

$$\text{Var}(\underline{z}^T A \underline{z}) = 2 \text{trace}((\Sigma^{1/2} A \Sigma^{1/2})^2)$$

$$= 2 \text{trace}(\Sigma^{1/2} A \Sigma^{1/2} \Sigma^{1/2} A \Sigma^{1/2})$$

$$= 2 \text{trace}(A \Sigma^{1/2} \Sigma^{1/2} A \Sigma^{1/2} \Sigma^{1/2})$$

$$= 2 \text{trace}(A \Sigma A \Sigma) = 2 \text{trace}((A \Sigma)^2).$$

$$\underline{\text{Ex 3}} \quad (a) \quad Z \sim N(\mu, 1) \Rightarrow Z - \mu \sim N(0, 1) \Rightarrow (Z - \mu)^2 \sim \chi^2_1$$

$$\Rightarrow E((Z - \mu)^2) = 1$$

$$\text{Now, } E(Z^2) = E((Z - \mu)^2) + \mu^2 + 2\mu \underbrace{E(Z - \mu)}_0 = 1 + \mu^2$$

$$\text{Similarly, } \text{Var}((Z - \mu)^2) = 2 \Rightarrow E((Z - \mu)^4) = 2 + 1 = 3$$

$$\left[\text{Also, for } Z' \sim N(0, \sigma^2) \right. \\ \left. E(Z'^4) = 3\sigma^2 \right]$$

$$\text{Now, } E(Z^4) = E((Z - \mu + \mu)^4)$$

$$= E((Z - \mu)^4 + 4(Z - \mu)^3\mu + 6(Z - \mu)^2\mu^2 + 4(Z - \mu)\mu^3 + \mu^4)$$

$$= 3 + 4 \times 0 + 6\mu^2 + 4 \times 0 + \mu^4 \quad \left[Z - \mu \sim N(0, 1), \text{ so all odd moments are 0} \right]$$

$$= 3 + 6\mu^2 + \mu^4$$

$$\text{Var}(Z^2) = E(Z^4) - \{E(Z^2)\}^2$$

$$= 3 + 6\mu^2 + \mu^4 - (1 + \mu^2)^2$$

$$= 3 + 6\mu^2 + \mu^4 - 1 - \mu^4 - 2\mu^2$$

$$= 2 + 4\mu^2$$

$$(b) \quad S \sim \chi^2_p(\delta) \Leftrightarrow S \stackrel{d}{=} \sum_{i=1}^p Z_i^2, \text{ where } Z_i \stackrel{\text{indep}}{\sim} N(\mu_i, 1) \\ \text{with } \sum_{i=1}^p \mu_i^2 = \delta$$

$$\text{Now, } E(S) = \sum_{i=1}^p E(Z_i^2)$$

$$= \sum_{i=1}^p (1 + \mu_i^2) = p + \sum_{i=1}^p \mu_i^2 = p + \delta$$

$$\text{Var}(S) = \sum_{i=1}^p \text{Var}(Z_i^2) \quad \left[\because Z_i \text{'s are indep.} \Rightarrow \text{Cov}(Z_i^2, Z_j^2) = 0 \forall i \neq j \right]$$

$$= \sum_{i=1}^p (2 + 4\mu_i^2) = 2p + 4\delta$$

For $S' \sim \chi^2_p$, (i.e., with $\delta = 0$), we get

$$E(S') = p \quad \text{and} \quad \text{Var}(S') = 2p$$

$$\therefore E(S) \geq E(S'), \quad \text{Var}(S) \geq \text{Var}(S'), \quad "=" \text{ iff } \delta = 0.$$

(c) $\underline{Z} \sim N_p(\underline{\mu}, I)$, $A \succeq 0$.

As before, use the spectral decomposition: $A = U \Lambda U^T = \sum_{i=1}^p \lambda_i \underline{u}_i \underline{u}_i^T$

$\underline{Z}^T A \underline{Z}$ becomes $\sum_{i=1}^p \lambda_i (\underline{u}_i^T \underline{Z})^2 = \sum_{i=1}^p \lambda_i w_i^2$

$\begin{pmatrix} w_1 \\ \vdots \\ w_p \end{pmatrix} = \begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_p^T \end{pmatrix} \underline{Z} \sim N_p(U \underline{\mu}, I)$, $U = \begin{pmatrix} \underline{u}_1^T \\ \vdots \\ \underline{u}_p^T \end{pmatrix}$ and $U U^T = I_p$

$\therefore w_i$'s are independent, $w_i \sim N(\underline{u}_i^T \underline{\mu}, 1)$

$\Rightarrow w_i^2$'s are independent, $w_i^2 \sim \chi_1^2((\underline{u}_i^T \underline{\mu})^2)$

Now, $E(\underline{Z}^T A \underline{Z}) = \sum_{i=1}^p \lambda_i E(w_i^2) = \sum_{i=1}^p \lambda_i (1 + (\underline{u}_i^T \underline{\mu})^2)$

$= \sum_{i=1}^p \lambda_i + \sum_{i=1}^p \lambda_i (\underline{u}_i^T \underline{\mu})^2$

$= \sum_{i=1}^p \lambda_i + \underline{\mu}^T \left(\sum_{i=1}^p \lambda_i \underline{u}_i \underline{u}_i^T \right) \underline{\mu}$

$= \text{trace}(A) + \underline{\mu}^T A \underline{\mu}$

$\text{Var}(\underline{Z}^T A \underline{Z}) = \sum_{i=1}^p \lambda_i^2 \text{Var}(w_i^2) = \sum_{i=1}^p \lambda_i^2 (2 + 4(\underline{u}_i^T \underline{\mu})^2)$

$= 2 \sum_{i=1}^p \lambda_i^2 + 4 \underline{\mu}^T \left(\sum_{i=1}^p \lambda_i^2 \underline{u}_i \underline{u}_i^T \right) \underline{\mu}$

$= 2 \text{trace}(A^2) + 4 \underline{\mu}^T A^2 \underline{\mu} \quad [A = U \Lambda U^T \Rightarrow A^2 = U \Lambda^2 U^T]$

(d) $\underline{Z} \sim N_p(\underline{\mu}, \Sigma)$, $\Sigma \succ 0 \Rightarrow \underline{Z}_0 = \Sigma^{-1/2} \underline{Z} \sim N_p(\Sigma^{-1/2} \underline{\mu}, I)$

$\underline{Z}^T A \underline{Z} = \underline{Z}_0^T \Sigma^{1/2} A \Sigma^{1/2} \underline{Z}_0 \sim \chi_r^2(\delta)$ if $\Sigma^{1/2} A \Sigma^{1/2}$ is idempotent $(\Rightarrow \Sigma^{1/2} A \Sigma^{1/2} \Sigma^{1/2} A \Sigma^{1/2} = \Sigma^{1/2} A \Sigma^{1/2})$

$(\Rightarrow A \Sigma A = A)$

In this case, $r = \text{trace}(\Sigma^{1/2} A \Sigma^{1/2}) = \text{trace}(A \Sigma)$

and $\delta = \underline{\mu}_0^T \Sigma^{1/2} A \Sigma^{1/2} \underline{\mu}_0$, where $\underline{\mu}_0 = E(\underline{Z}_0) = \Sigma^{-1/2} \underline{\mu}$
 $= \underline{\mu}^T A \underline{\mu}$.

(e) Take $A = \Sigma^{-1}$. Then, we get $A \Sigma A = \Sigma^{-1} \Sigma \Sigma^{-1} = \Sigma^{-1} = A$

Also, $\text{tr}(A \Sigma) = \text{tr}(\Sigma^{-1} \Sigma) = \text{tr}(I_p) = p$

$\therefore \underline{z}^T \Sigma^{-1} \underline{z} \sim \chi_p^2(\delta)$, where $\delta = \underline{\mu}^T \Sigma^{-1} \underline{\mu}$.

(f) Similar to part (d), $\underline{z}^T A \underline{z} = \underline{z}_0^T \Sigma^{1/2} A \Sigma^{1/2} \underline{z}_0$, where $\underline{z}_0 \sim N_p(\Sigma^{-1/2} \underline{\mu}, I)$

$\therefore$ From (c), $E(\underline{z}^T A \underline{z}) = \text{tr}(\Sigma^{1/2} A \Sigma^{1/2}) + \underline{\mu}^T \Sigma^{-1/2} \Sigma^{1/2} A \Sigma^{1/2} \Sigma^{-1/2} \underline{\mu}$
 $= \text{tr}(A \Sigma) + \underline{\mu}^T A \underline{\mu}$

Again, from (c), $\text{Var}(\underline{z}^T A \underline{z}) = 2 \text{tr}((\Sigma^{1/2} A \Sigma^{1/2})^2) + 4 \underline{\mu}^T \Sigma^{-1/2} (\Sigma^{1/2} A \Sigma^{1/2})^2 \Sigma^{-1/2} \underline{\mu}$
 $= 2 \text{tr}(A \Sigma)^2 + 4 \underline{\mu}^T A \Sigma A \underline{\mu}$

Ex 4 $\hat{\underline{\theta}}_{LSE} \sim N_d(\underline{\theta}, \sigma^2 (X^T X)^{-1})$
 $\frac{SSE}{\sigma^2} \sim \chi_{n-d}^2$ } indep.

(a) $\underline{a}^T \hat{\underline{\theta}}_{LSE} \sim N(\underline{a}^T \underline{\theta}, \sigma^2 \underline{a}^T (X^T X)^{-1} \underline{a})$
 $\frac{SSE}{\sigma^2} \sim \chi_{n-d}^2$ } indep.

$\Rightarrow T_\theta = \frac{\underline{a}^T \hat{\underline{\theta}}_{LSE} - \underline{a}^T \underline{\theta}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}} \sqrt{\frac{SSE}{n-d}}} \sim t_{n-d}$

So, T_θ is a pivot. We can find a C.I. for $\underline{a}^T \underline{\theta}$ by inverting the pivot.

$IP_\theta \left[-t_{\alpha/2; n-d} \leq T_\theta \leq t_{\alpha/2; n-d} \right] = 1 - \alpha$
 $t_{\alpha; \nu}$ is the upper α -quantile of t_ν dist.

$$\Leftrightarrow P_{\theta} \left[-t_{\alpha/2; n-d} \leq \frac{\underline{a}^T \hat{\underline{\theta}}_{LSE} - \underline{a}^T \underline{\theta}}{\sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}} \sqrt{\frac{SSE}{n-d}}} \leq t_{\alpha/2; n-d} \right] = 1-\alpha$$

$$\Leftrightarrow P_{\theta} \left[\begin{aligned} &\underline{a}^T \hat{\underline{\theta}}_{LSE} - \sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}} \sqrt{\frac{SSE}{n-d}} t_{\alpha/2; n-d} \\ &\leq \underline{a}^T \underline{\theta} \\ &\leq \underline{a}^T \hat{\underline{\theta}}_{LSE} + \sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}} \sqrt{\frac{SSE}{n-d}} t_{\alpha/2; n-d} \end{aligned} \right] = 1-\alpha$$

So, an $100(1-\alpha)\%$ CI for $\underline{a}^T \underline{\theta}$ is

$$\left[\underline{a}^T \hat{\underline{\theta}}_{LSE} \pm \sqrt{\underline{a}^T (X^T X)^{-1} \underline{a}} \sqrt{\frac{SSE}{n-d}} t_{\alpha/2; n-d} \right]$$

$$(b) \quad y = \underline{x}^T \underline{\theta} + \varepsilon$$

$$\Rightarrow E(y) = \underline{x}^T \underline{\theta}$$

We can predict y_{new} at $\underline{x}_{new}$ by $\hat{y}_{new} = \underline{x}_{new}^T \hat{\underline{\theta}}_{LSE}$.

(c) It may be tempting to use

$$\left[\underline{x}_{new}^T \hat{\underline{\theta}}_{LSE} \pm \sqrt{\underline{x}_{new}^T (X^T X)^{-1} \underline{x}_{new}} \sqrt{\frac{SSE}{n-d}} t_{\alpha/2; n-d} \right]$$

as the prediction interval. But remember

$$y_{new} = \underline{x}_{new}^T \underline{\theta} + \varepsilon$$

$$\underline{x}_{new}^T \hat{\underline{\theta}}_{LSE} \sim N(\underline{x}_{new}^T \underline{\theta}, \sigma^2 \underline{x}_{new}^T (X^T X)^{-1} \underline{x}_{new})$$

$$\therefore \underline{x}_{new}^T \hat{\underline{\theta}}_{LSE} + \varepsilon \sim N(\underline{x}_{new}^T \underline{\theta}, \sigma^2 (\underline{x}_{new}^T (X^T X)^{-1} \underline{x}_{new} + 1))$$

This gives us a prediction interval from

$$\underline{x}_{\text{new}}^T \hat{\underline{\theta}}_{\text{LSE}} - y_{\text{new}} = \underline{x}_{\text{new}}^T (\hat{\underline{\theta}}_{\text{LSE}} - \underline{\theta}) - \varepsilon$$
$$\sim N(0, \sigma^2 \{1 + \underline{x}_{\text{new}}^T (\underline{X}^T \underline{X})^{-1} \underline{x}_{\text{new}}\})$$
$$\frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-1} \quad \text{indep.}$$

$$\Rightarrow \frac{\underline{x}_{\text{new}}^T \hat{\underline{\theta}}_{\text{LSE}} - y_{\text{new}}}{\sqrt{1 + \underline{x}_{\text{new}}^T (\underline{X}^T \underline{X})^{-1} \underline{x}_{\text{new}}}} \sqrt{\frac{\text{SSE}}{n-1}} \sim t_{n-d}$$

Inverting this, we get the following prediction interval for y_{new}

$$\left[\underline{x}_{\text{new}}^T \hat{\underline{\theta}}_{\text{LS}} \mp \sqrt{1 + \underline{x}_{\text{new}}^T (\underline{X}^T \underline{X})^{-1} \underline{x}_{\text{new}}} \sqrt{\frac{\text{SSE}}{n-d}} t_{d/2, n-d} \right]$$

Notice that the prediction interval is wider than the usual confidence interval. This is due to the additional uncertainty from ε in the model.